Empty Quarter MAGs — End-to-End Pipeline Guide

Forensic reconstruction: every stage, tool, command and output path — from raw reads to the three-pillar story

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Contents

0. How to read this guide **2**

1. Pipeline map (one page) **3**

2. Data & sampling **4**

3. Stage-by-stage (forensic) **5**

3.1 Read QC — fastp (Snakemake rule `qc_reads`) . . . 5
3.2 Assembly — MEGAHIT (Snakemake rules `megahit` → `rename_contigs` → `quast`) . . . 6
3.3 Binning — SemiBin2: SELF-SUPERVISED (project pipeline) vs PRETRAINED (Robert's re-bin) ⚠⚠ . . . 6
3.4 MAG QC — CheckM2 . . . 7
3.5 Annotation — Bakta + eggNOG-mapper . . . 8
3.6 Dereplication — dRep ⚠ (read the three-counts box) . . . 9
3.7 Taxonomy — GTDB-Tk (defines the census; Subversion 1) . . . 10
3.8 Protein catalogue — MMseqs2 linclust ⚠ (two catalogues exist) . . . 10
3.9 Sequence dark matter — UniRef50 search . . . 11
3.10 Structural dark matter — ProstT5 (3Di) + FoldSeek vs AFDB . . . 11
3.11 Novel families (the 334,810) — 30%-identity sequence clustering . . . 12
3.12 Novel folds (the 68,808 + 68,527) — fold clustering + ESMFold . . . 12
3.13 Part A — A1 atmospheric "life on air" ⚠ (DIAMOND, not HMMER) . . . 13
3.14 Part A — A2 UV/desiccation repair . . . 13
3.15 Part A — A3 osmolytes (dark-family vignette) . . . 14
3.16 Part A — A4 BGCs + rhodopsins + (alt-nitrogenase NULL) . . . 14
3.17 Part A — A5 resistome / biosafety . . . 15
3.18 Part A — A-general marker census (eggNOG KO → GTDB phylum) . . . 15
3.19 Part B — B2 selection (dN/dS) ⚠ (Biopython NG86, not codeml) . . . 16
3.20 Part B — B3 ecology (CoverM + bespoke PERMANOVA) ⚠ (not vegan) . . . 16
3.21 Part B — B3d abundance (not a contamination tail) . . . 17
3.22 Part B — B4 diversifying selection + Boltz-2 . . . 17
3.23 Part B — B5 decode (C·V·T) ⚠ (dossier = 73, not 10) . . . 17
3.24 Part C — C1 PICRUSt2 reference bias . . . 18
3.25 Part C — C1a pangenome openness ⚠ (ran 2026-06-28; scipy missing) . . . 18
3.26 Part C — C2 SMETANA community (16-community pilot) ⚠ (detailed deferred) 18

3.27 Part C — C3 global / desert specificity ⚠ (SPIRE not landed) 19

4. Three pillars → results map **19**

5. Reproducibility & setup gaps **20**

6. Parked / not-adopted tracks (so nothing is mistaken for the main path) **21**

7. The catalogue/binning issue & remediation plan ⚠⚠ (read this) **22**
7.1 The issue in one paragraph 22
7.2 What you actually have: a best-of-both MERGE 22
7.3 What the merge HAS vs what is MISSING 23
7.4 How to fix it — staged plan (do the cheap, decision-informing steps first) . . . 24

8. Glossary & naming conventions **25**

0. How to read this guide

This document reconstructs the **entire** Empty Quarter (Rub' al-Khali) MAGs/proteome pipeline from the *real* compute artefacts on IBEX — not from the paper's curated narrative.

> ⚠⚠ **Most important finding — read §7 first.** The catalogue was built from the **pretrained** single-sample bins, not the intended **self-supervised** ones; a best-of-both merged catalogue (9,211 reps) exists but was never carried downstream. §7 gives the full have/missing breakdown and a staged fix plan. Every stage below was verified by reading the actual driver scripts, the Slurm logs, and the on-disk outputs; **counts are quoted with the command that produced them**. Where the curated narrative and the artefacts disagree, the artefact wins and the discrepancy is flagged with ⚠.

The two trees you need to know

Name	Path	What it holds
Paper repo (`$REPO`)	`alelopezv/repos/empty-quarter-mags`	manuscript, curated `results/*.tsv`, paper-specific `code/`, this guide
Workbench (`$WB`)	`rob/metagenomics`	the heavy pipeline: all stage dirs (`01_qc` … `09_*`), `scripts/`, `scripts/ibex/`, `logs/`, `databases/`
Data root	`EmptyQuarter_Data`	raw reads, assemblies, per-sample binning/annotation tree, large-file staging

All paths in this guide are written **relative to the project root** `/ibex/scratch/projects/c2014/` (e.g. `rob/metagenomics/...`, `EmptyQuarter_Data/...`); a few data dirs are also mirrored under `/ibex/project/c2014/`.

> ⚠ **The `$REPO/workbench` symlink is broken** — it points at `/home/leechuck/Public/software/empty-quarter-workbench`, which does not resolve on this machine. Use the absolute `$WB` path above. (See §5 for the fix.)

Conventions

- **Compute model:** everything heavy runs through Slurm `sbatch` on IBEX; account `pi-hohndor`; job-name prefix `hohndor_`. GPU work (ProstT5, ESMFold) ran on IBEX A100s and an external **ORIX** cluster (H200) used as free-cycle overflow. "Local python" = scripts in `$REPO/code/` run on a workstation, not Slurm.
- **Tool versions** were resolved from the conda environments under `rob/miniconda3/envs/*/conda-meta/` when not pinned in the script. Versions taken this way are marked *(conda-meta)*.
- **MAG ID format:** `<sample>__SemiBin_<n>` (e.g. `10Dr2__SemiBin_0`). The `__SemiBin_` infix confirms a bin is a **SemiBin2** product (not DAS_Tool).
- **Sample ID decoding:** `<site><compartment><trip>` — e.g. `11PRr3` = site 11, compartment `PR` (PlantRoot/rhizosphere), trip `r3`. Compartments: `D`/`S` = Surface/Deep soil tiers, `PR` = PlantRoot. Trips `r1`…`r3` (and `F`-prefixed re-runs).

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1. Pipeline map (one page)

```
EQ_WGS Snakemake   (alelopezv/EQ_WGS)
=====================================================================
RAW READS  (Illumina PE)
   |  fastp 1.0.1    phred20, len>=50, complexity>=20; no host removal
   v
QC READS
   |  MEGAHIT 1.2.9  meta-large, min-contig 500, min-count 2  -> rename -> QUAST
   v
ASSEMBLIES (572)   wgs_assemblies/assemblies/<sample>/<sample>_contigs.fa
   |
   +- Track 1  SELF-SUPERVISED   (Snakemake - the project pipeline)
   |     BWA-MEM (paired+SE) -> SemiBin2 --self-supervised -m 1000
   |     -> wgs_assemblies/binning/<sample>/semibin/output_bins/
   |
   +- Track 2  PRETRAINED   (Robert's separate re-bin)
         minimap2 -> SemiBin2 --environment global  (resolves to -m 2500)
         -> .../processed/.../<sample>/wgs/binning/single_sample/output_bins/
```

```
   v
CheckM2 1.1.0  -> HQ / MQ / LQ
   |  single_sample 15,469   .   self_supervised 21,616      (HQ+MQ)
   |  [PARKED] DAS_Tool consensus -> 5,438 reps   (not the catalogue; see 6)
   v
CATALOGUE = 15,469 single_sample HQ+MQ MAGs        (pretrained; see 7)
   |  Bakta 1.12.0  +  eggNOG 2.1.13      (gene calls, KO / orthology)
   v
GTDB-Tk 2.4.1 / r226   ->  44 phyla, 0 novel          (Subversion 1)
   v
PROTEIN CATALOGUE   MMseqs2 linclust  51.9M prot -> 23.4M reps
   |  search vs UniRef50 2026_01  ->  2.84M "dark"  (12.1%)
   |
   +- PART A  (the readable proteome)
   |    A1 atmospheric   DIAMOND/HydDB + coxL form-I + MAFFT/FastTree trees
   |    A2 UV/desicc     eggNOG 28-KO panel
   |    A3 osmolytes     dark-family osmolyte-EC vignette
   |    A4 BGCs          antiSMASH 8.0.4 --minimal; rhodopsins; vnf/anf null
   |    A5 resistome     AMRFinderPlus 4.2.7; VFDB; pathogen census
   |
   +- PART B  (the dark proteome)
   |    ProstT5 3Di + FoldSeek vs AFDB  ->  2.53M struct-novel (89.24%)
   |    30%-id clustering -> 334,810 novel families
   |    tier1 folds -> 68,808 novel-fold families; ESMFold -> 68,527 structures
   |    B2 dN/dS (NG86) 0.391 vs 0.286     B3 endemism 45.6% vs 2.3%
   |    B4 FUBAR + Boltz-2 (dark lectin)   B5 C.V.T decode (top-10)
   |
   +- PART C  (the bridge)
        C1   PICRUSt2 vs MAG KO (436): 2.4x over-predict, 56.7% no-KO
        C1a  pangenome openness: 337/345 species open
        C2   gapseq+SMETANA (16): MRO ~0.69, guild-feeder refuted
        C3   vs MGnify90 / GEM / 5 deserts: 21.4% homolog-free anywhere
```

---

2. Data & sampling

- **System:** Rub’ al-Khali (“Empty Quarter”) hyperarid sand sea. Design ≈ 60 sites × 5 trips × compartments (Surface / Deep / PlantRoot-rhizosphere), with XRF geochemistry, climate, and depth metadata.
- **Sample counts seen in the pipeline** (these differ by stage because not every sample binned/assembled/recruited):
  - **572** sample assemblies (`find .../wgs_assemblies/assemblies -name '*_contigs.fa' |`

wc -l).
  - **550** binnable samples (metadata/binnable_samples.txt).
  - **551** samples in the dark-family ecology read set (darkfam_ecology/canon_samples.txt).
  - **547** samples with full community+environment (used by PERMANOVA).
  - **436** samples matched for the PICRUSt2 comparison (both 16S and MAGs).
- **Raw reads:** alelopezv/230611_phages/dsDNA/EQ_WGS_phabox/...Lane*/
- **QC reads:** EmptyQuarter_Data/soil/wgs_assemblies/processed_reads/<sample>_R{1,2}.fastq.gz
- **Assemblies:** EmptyQuarter_Data/soil/wgs_assemblies/assemblies/<sample>/<sample>_contigs.fa
- **Per-sample binning/annotation tree:** EmptyQuarter_Data/processed/sequencing-results/sites/<trip>/site<N>/<sample>/wgs/{binning,quality,annotation}/...
- **XRF geochemistry table:** xrf_lab_table_all_trips.tsv (joined into env_matrix.tsv by $REPO/code/build_env_matrix.py).

---

3. Stage-by-stage (forensic)

Each stage uses the same template: **Purpose · Tool+version · Env · Command · Inputs · Outputs+counts · Job IDs/logs · Caveats**.

> **UPSTREAM PROVENANCE (§3.1–3.3).** QC, assembly and the **authoritative (self-supervised)** binning were run from a **Snakemake workflow** at alelopezv/EQ_WGS (assembly.smk, binning.smk, rules/, config/config.yml, per-rule conda envs/*.yml), writing to …/EmptyQuarter_Data/soil/wgs_assemblies/. This is the source of truth for the upstream commands below — superseding the standalone *.sh templates in wgs_assemblies/ (which are stale). The **pretrained** single_sample binning (§3.3) is a *separate re-binning Robert ran* in the metagenomics workbench, not part of this Snakemake.

3.1 Read QC — fastp (Snakemake rule `qc_reads`)

- **Purpose:** adapter/quality trim, base correction, low-complexity filter before assembly.
- **Tool + version:** fastp **1.0.1** *(conda-meta; env `EQ_WGS/envs/fastp.yml`)*.
- **Command** (EQ_WGS/rules/assembly/qc_reads.rule; params from config.yml):

```
fastp --thread 8 -i <R1> -I <R2> -o <out_R1> -O <out_R2> \
      --unpaired1 <U1> --unpaired2 <U2> -h <html> -j <json> \
      --correction --qualified_quality_phred 20 --unqualified_percent_limit 40 \
      --length_required 50 --complexity_threshold 20
cat <U1> <U2> > <SE>.fastq.gz        # unpaired reads merged into a single-end file
```

- **Inputs:** raw lane FASTQs (all_samples.tsv cols R1_raw/R2_raw). **Outputs:**

wgs_assemblies/processed_reads/<sample>_{R1,R2,SE}.fastq.gz + report.{html,json}.

- **Logs:** wgs_assemblies/logs/fastp/<sample>.log (Snakemake per-rule).
- **Caveats:** **No host/eukaryote read removal** anywhere in the pipeline (no DeepMicroClass / no prokaryote pre-extraction); --complexity_threshold 20 is the only contaminant-style filter. The SemiBin2 rule (§3.3) bins the **assembly contigs directly**.

3.2 Assembly — MEGAHIT (Snakemake rules megahit → rename_contigs → quast)

- **Purpose:** de novo per-sample assembly, contig rename, assembly QC.
- **Tool + version:** MEGAHIT **v1.2.9** (env megahit.yml); QUAST **5.2.0** (env quast.yml).
- **Command** (EQ_WGS/rules/assembly/assembly.rule; params from config.yml):

```
megahit -1 <R1> -2 <R2> -r <SE> -o assemblies/<sample> \
        --presets meta-large --min-contig-len 500 --min-count 2 \
        --low-local-ratio 0.3 --prune-level 1 --num-cpu-threads 16
# rename_contigs.rule:  sed 's/^>[^_]*_\(...\)/>{sample}_\1/' final.contigs.fa >
↪  <sample>_contigs.fa
# quast.rule:           quast.py <sample>_contigs.fa -o assemblies/qc/<sample>
```

- **Outputs:** wgs_assemblies/assemblies/<sample>/<sample>_contigs.fa; QC assemblies/qc/<sample>/report.txt. **Count:** find … -name '*_contigs.fa' | wc -l → **572**.
- **Logs:** wgs_assemblies/logs/{megahit,rename_contigs,quast}/<sample>.log.
- **Note:** min_count: 2 is now **confirmed** from config.yml (resolves the earlier .sh-vs-options.json discrepancy — 2 is correct).

3.3 Binning — SemiBin2: SELF-SUPERVISED (project pipeline) vs PRETRAINED (Robert's re-bin) ⚠⚠

- **Purpose:** recover MAGs. **Two SemiBin2 binnings exist per sample, by two different methods.** Which one fed the catalogue is the central issue (→ §7).
- **Tool + version:** SemiBin2 **2.2.1**.

Track 1 — SELF-SUPERVISED (the authoritative project binning; Snakemake). - **Coverage** (EQ_WGS/rules/binning/run_bwa.rule, env bwa_env.yml): **BWA-MEM**, paired **and** SE reads, merged + sorted → <sample>_metaG.reads.sorted.bam (note: includes SE). - **Binning** (EQ_WGS/rules/binning/semibin.rule, env semibin2_env.yml): bash SemiBin2 single_easy_bin --tmpdir /tmp -m 1000 \ --input-fasta <sample>_contigs.fa --input-bam <sample>_metaG.reads.sorted.bam \ --self-supervised --output <sample>/semibin --threads 24 ⇒ **genuinely self-supervised** (--self-supervised → trains a per-sample model.pt, log "Training model..."), **min-contig 1000** (-m 1000, config semibin.min_len). - **Output:** wgs_assemblies/binning/<sample>/semibin/output_bins/SemiBin_*.fa.

Track 2 — PRETRAINED global (a separate re-binning Robert ran in the work-

bench). - **Coverage + binning** (`$WB/scripts/01_semibin2_single.sh`): `bash minimap2 -a -x sr -t 16 $CONTIGS $R1 $R2 | samtools sort -o $BAM # paired only, no SE SemiBin2 single_easy_bin -i $CONTIGS -b $BAM -o $BINDIR \ --environment global --sequencing-type short_read -t 16` ⇒ **pretrained `global` model** (no training, no `model.pt`, log "Generating training data" only), **`-m` not set** → `min_len=None, ratio=0.05` → SemiBin2 adaptive rule resolves to **2500 bp** for these fragmented soil assemblies (verified 20/20; see §7.2) — i.e. it binned only contigs ≥2500 bp, dropping ~27% of each assembly that self-sup (`-m 1000`) kept. - **Output:** `…/processed/sequencing-results/sites/<trip>/site<N>/<sample>/wgs/binning/single_sample/output_bins/SemiBin_*.fa.gz`. - **Note:** Robert also gzipped **copies** of the self-supervised bins into `…/wgs/binning/self_supervised/output_bins/` (`01b_self_supervised_gzip_array.sh`) for his dRep.

> ⚠⚠ **THE CENTRAL ISSUE.** The catalogue (`03_annotation/mag_list.txt`, 15,469) is built entirely from **Track 2 (pretrained)**; the entire downstream pipeline ran on it. The **self-supervised** Track 1 — your project pipeline's intended output, which recovers more and better bins — only fed the unadopted `04_drep_self_vs_single` comparison. The two tracks also differ in coverage mapper (BWA+SE vs minimap2 paired-only) and min-contig (1000 vs default). **Full have/missing state + fix plan: §7.**

3.4 MAG QC — CheckM2

- **Purpose:** completeness/contamination → HQ/MQ/LQ tiers (MIMAG-style).
- **Tool + version:** CheckM2 **1.1.0** *(conda-meta)*; DB `databases/checkm2/.../uniref100.KO.1.dmnd`.
- **Env:** `activate checkm2`; `export CHECKM2DB=….`
- **Command** (`$WB/scripts/02_checkm2_single.sh`, `02b_…self_supervised.sh`):

```
checkm2 predict --input $TMP --output-directory $QCDIR -x .fa --threads 8 \
    --database_path $CHECKM2DB --force
```

- **Outputs / counts** — authoritative CheckM2 class breakdown (from the aggregated tables `results/checkm2_{single_sample,self_supervised}_full.tsv`; columns `sample · bin · fa_path · completeness · contamination · class`):

Track	Total bins	HQ	MQ	LQ	HQ+MQ
single_sample (pretrained)	45,869	3,842	11,635	30,392	15,477
self_supervised	86,079	4,507	17,109	64,463	21,616

  - Per-sample reports (both tracks): `.../<sample>/wgs/quality/{single_sample,self_supervised}/qual` (+ `busco`, `protein_files`, `diamond_output`).
  - Combined both-track completeness/contamination table (the dRep input): `04_drep_self_vs_single/genomeInfo.csv` (37,085 rows).

- **Caveats:** ⚠ the **catalogue** manifest `03_annotation/mag_list.txt` = **15,469** single_sample bins — **8 fewer** than CheckM2's 15,477 single HQ+MQ. The 8 are a **contamination-boundary** effect: all sit at contamination **exactly 10.00**, so the catalogue (MIMAG `< 10`, strict) drops them while the aggregated `class` column (`≤ 10`) keeps them as MQ (25Sr1/SemiBin_2, 28Dr3/SemiBin_18, 34Dr1/SemiBin_3, F13PRr2/SemiBin_34, F26PRr1/SemiBin_84, F49PRr3/SemiBin_13, V14PRr1/SemiBin_56, V35PRr2/SemiBin_18). So 15,469 is the MIMAG-correct count; 15,477 is marginally lenient. The catalogue is a clean strict subset (0 bins in catalogue but not CheckM2 HQ+MQ). Self-sup HQ+MQ matches (21,616).
- **Thresholds** (`03b_aggregate_self_supervised_quality.py::classify()`): **HQ = completeness ≥90 & contamination ≤5; MQ = ≥50 & ≤10; LQ otherwise** (MIMAG completeness/contamination cutoffs — rRNA/tRNA criteria are **not** enforced). `classify()` uses contamination **≤10**, so it (and the self-sup list) keep the 8 bins at cont = 10.00; the single_sample catalogue manifest was built earlier with a strict **<10**, which is exactly why it has those 8 fewer.

3.5 Annotation — Bakta + eggNOG-mapper

- **Purpose:** gene calling + structural annotation (Bakta), then functional/orthology (eggNOG).
- ⚠ **Scope differs by track** (verified per-sample): **single_sample annotated HQ+MQ only** (15,469); **self_supervised annotated ALL bins incl. LQ** (86,079). E.g. 59PRr1: single 66/143 bins (= its HQ+MQ) vs self-sup 218/218 (= all).
- **Tools:** Bakta **1.12.0** (DB **v6.0**); eggNOG-mapper **2.1.13** (DB dir labelled `2.1.12`).
- **Commands:**

```
bakta --db $BAKTA_DB --output $OUTDIR --prefix $BIN --threads 8 --skip-plot --force $FA
↪   # 03_bakta_array.sh
emapper.py -i $PROTEINS --output_dir $OUTDIR --data_dir $EGGNOG_DB --cpu 16 \
    --itype proteins --override                                                    #
    ↪  04_eggnog_array.sh
```

- **Inputs:** single — `03_annotation/mag_list.txt` (15,469 HQ+MQ); self-sup — `mag_list_self_supervised_all.txt` (86,079, **all** bins).
- **Outputs:** per-MAG, both tracks: `.../<sample>/wgs/annotation/{single_sample,self_supervised}/{bakta` (the `.emapper.annotations` power A2/A4/A5/C1).
- **Caveats:** ⚠ the protein catalogue (§3.8) pooled **only the single_sample (HQ+MQ) annotation** (15,469 `.faa`); the self-sup annotation (all bins, incl. LQ) exists but was **not** pooled downstream. Bakta tool 1.12.0 vs DB 6.0; eggNOG tool 2.1.13 vs DB-dir 2.1.12 (label mismatch).

3.6 Dereplication — dRep ⚠ (read the three-counts box)

- **Purpose:** species-level (95% ANI) dereplication of pooled single + self_supervised HQ+MQ MAGs.
- **Tool + version:** dRep **3.6.2** (`dRep version 3.6.2 was run`); fastANI **1.34**.
- **Env:** `activate drep`.
- **Command** (`$WB/scripts/05b_drep_run.sh`; prep `05_drep_prep.sh` symlinks inputs as `ss__…` / `selfsup__…`):

```
dRep dereplicate $OUTDIR -g $INPUTS/*.fa.gz --genomeInfo $GENOMEINFO \
    --S_algorithm fastANI -comp 50 -con 10 -pa 0.9 -sa 0.95 -p 32
```

- **Inputs:** `04_drep_self_vs_single/inputs/` = **37,085** prefixed symlinks.
- **Outputs:** `04_drep_self_vs_single/out/dereplicated_genomes/` = **9,211** reps (`ls | wc -l`). Winner provenance (`Wdb.csv`): **selfsup 6,308 / single 2,903**. Post-hoc novelty split: `new_mags.tsv` 3,387, `redundant_mags.tsv` 18,231.

⚠ **THREE DIFFERENT "REPRESENTATIVE" COUNTS — do not conflate:**

Count	What it is	Where
15,469	the **catalogue** = single_sample HQ+MQ MAGs; the GTDB-Tk input; the paper's headline	`03_annotation/mag_list.txt`, `02_taxonomy/gtdbtk_out/`
9,211	dRep species reps of the **pooled single+self** HQ+MQ set (a single-vs-self *comparison*)	`04_drep_self_vs_single/out/`
5,438	dRep reps of the **DAS_Tool-refined** bins (parked experiment, §6)	`05_refine/_drep/out/`

The **15,469** figure is the only one that exactly matches `mag_list.txt` and the GTDB summaries, and its MAGs carry `__SemiBin_` IDs ⇒ the catalogue is **single_sample SemiBin2 HQ+MQ (pretrained `--environment global`)**, and is **not** the cross-sample-dereplicated 9,211 set.

⚠⚠ **Load-bearing finding:** in the 9,211-rep `04_drep_self_vs_single` dereplication, the **self-supervised bins WON 6,308 of 9,211 species reps (68%)** vs 2,903 for single_sample (`Wdb.csv`). So the dRep step that *did* incorporate the (correct, self-supervised) bins was **computed and then NOT carried forward** — the downstream pipeline pulled `mag_list.txt` (single_sample / pretrained) instead. Per-sample the self-sup track also recovers more HQ+MQ bins (e.g. 59PRr1: 81 self-sup vs 66 single_sample). **Net: the self-supervised work was done,**

validated, and dominated dereplication, but the catalogue + all downstream analyses ran on the pretrained single_sample set. Escalate to Robert — the catalogue may need rebuilding on the self-sup / 9,211-rep set. This is the single biggest provenance issue in the pipeline.

3.7 Taxonomy — GTDB-Tk (defines the census; Subversion 1)

- **Purpose:** GTDB classification of the catalogue; the "tree is ordinary" result.
- **Tool + version:** GTDB-Tk **v2.4.1**, reference **r226** (`/ibex/reference/KSL/gtdb-tk/release226`).
- **Env:** `activate gtdbtk`; `export GTDBTK_DATA_PATH=…/release226`.
- **Command** (`08_gtdbtk_classify_chunked.sh`, 16-way array; merged by `09_gtdbtk_merge.sh`):

```
gtdbtk classify_wf --genome_dir $MAG_DIR --out_dir $OUTDIR \
    --extension fa.gz --cpus 32 --pplacer_cpus 16 --skip_ani_screen
```

- **Inputs:** `02_taxonomy/gtdbtk_input/` = **15,469** `<sample>__<bin>.fa.gz` (== `mag_list.txt`).
- **Outputs / counts** (the FINAL catalogue summaries):
  - `gtdbtk.bac120.summary.tsv` → **14,752** (`tail -n+2 | wc -l`)
  - `gtdbtk.ar53.summary.tsv` → **717** ⇒ **14,752 + 717 = 15,469** ✔
  - Phyla = **44** (37 bac + 7 ar); Actinomycetota **6,788**; Pseudomonadota **3,740**; **0 novel phyla/classes**. *(These four are independently reproduced by `reproducibility/verify.py`.)*
- **Caveats:** the **chunked** path (`08_gtdbtk_classify_chunked.sh`) is the one used; `--skip_ani_screen` set.

3.8 Protein catalogue — MMseqs2 linclust ⚠ (two catalogues exist)

⚠ **There are TWO parallel protein catalogues.** The paper's headline ladder (51.9M → 23.4M → 2.84M → 2.53M → 334,810 → 68,808 → 68,527) is the **cluster90** path (Bakta proteins from MAG bins, dir `04_structure/`). A second, larger **delta** catalogue (pyrodigal on *all assemblies*, dir `04_protein_catalog/`: 753M → 414M reps → 68.8M dark) exists and feeds the tier1/tier2 structural fold-clustering. Which catalogue the paper calls "the catalogue" is a paper-repo decision; everything below is the cluster90 path unless noted.

- **Purpose:** pool all single-sample Bakta proteins, cluster at 90% identity → non-redundant reps.
- **Tool + version:** MMseqs2 **18.8cc5c** (`easy-linclust`), from the `foldseek` env.
- **Command** (`$WB/scripts/05_pool_and_cluster.sh`):

```
mmseqs easy-linclust $POOLED 04_structure/cluster90 $TMPDIR --min-seq-id 0.9 -c 0.8
↪  --cov-mode 0 --threads 32
```

- **Outputs / counts** (`logs/linclust_46528255.out`): pooled **51,920,778** proteins from 15,469 files → **23,416,070** representatives (`04_structure/cluster90_rep_seq.fasta`).
- **Delta companion:** pyrodigal **3.7.1** `-p meta` over 552 assemblies → 753,305,533 proteins → linclust90 → 414,574,825 reps (`04_protein_catalog/`, scripts `22_pyrodigal…`, `26_linclust_delta.sh`).

3.9 Sequence dark matter — UniRef50 search

- **Purpose:** reps with no UniRef50 hit are sequence-"dark".
- **Tool + version:** MMseqs2 18.8cc5c (`search` + `convertalis`). DB **UniRef50 Release 2026_01** (28-Jan-2026, 60,315,044 clusters).
- **Command** (array `$WB/scripts/06_seq_vs_uniref50_array.sh`):

```
mmseqs search $QUERYDB $UNIREF50_DB $OUT/result_aln $TMP --threads 16 -e 1e-3 -s 7.5
↪  --max-seqs 10
```

- **Outputs / counts** (`logs/uniref50_merge_46698934.out`): reps 23,416,070; hit 20,577,040; **nohit = 2,839,030** (~12.1%) → `04_structure/seq_vs_uniref50/nohit_reps.fasta`.
- **Delta companion:** `27_uniref50_search_delta.sh` (`--array=0-1199`) → nohit **68,798,040**.

3.10 Structural dark matter — ProstT5 (3Di) + FoldSeek vs AFDB

- **Purpose:** predict a 3Di structural alphabet for the 2.84M dark reps; FoldSeek vs AFDB; no-AFDB-hit = **structurally novel**.
- **Tool + version:** FoldSeek standalone GPU binary, commit `8dc75c74…` (conda foldseek pkg `10.941cd33`); ProstT5 weights `prostt5-f16.gguf` (GGML f16).
- **Env:** `module load rl9-gpustack cuda/12.4.1` (no conda; standalone binary). A100 required (GGML CUDA, cc≥8.0).
- **Command** (`06d_prostt5_array.sh`, `06g_…`):

```
$FOLDSEEK createdb $CHUNK_FA $QUERY_DB --prostt5-model $PROSTT5_MODEL --gpu 1 --threads
↪  16
$FOLDSEEK search  $QUERY_DB $AFDB $OUT/afdb_aln $TMP --threads 16 -e 1e-3 -s 9.5
↪  --db-load-mode 2 --split-memory-limit 80G
```

- **Outputs / counts** (`logs/orix_fmerge_46884756.out`): queries 2,839,030; afdb_hit 305,584 (10.76%); **struct_nohit = 2,533,446 = 89.24%** → `04_structure/foldseek_results/struct_nohit.fasta`.
- **Caveats:** load-balanced IBEX A100 + ORIX H200. ProstT5 3Di is a *predicted* alphabet — the genuinely-novel folds are re-confirmed with explicit ESMFold coordinates

(§3.12).

3.11 Novel families (the 334,810) — 30%-identity sequence clustering

- **Purpose:** cluster the 2.84M dark reps at family (30% id) level → recurrent novel families.
- **Tool + version:** MMseqs2 `easy-cluster` (foldseek env).
- **Command** (`scripts/analysis/novel_family_discovery.py` via `07_novelty_analysis.sh`):

```
mmseqs easy-cluster nohit_reps.fasta <prefix> $tmp --min-seq-id 0.3 -c 0.5 --cov-mode 0
↪  --threads 32
```

- **Outputs / counts** (`logs/novelty_46706193.out`): 1,875,540 families total; 1,540,730 singletons; **334,810 multi-member** novel families.
- **Caveats:** ⚠ **334,810 is a *sequence*-family count (30% id), NOT a fold count.** It pairs with — but is computed entirely separately from — the 68,808 fold families below. Never sum the two.

3.12 Novel folds (the 68,808 + 68,527) — fold clustering + ESMFold

- **Purpose:** structurally cluster the dark proteins into fold families; a "novel-fold family" = cluster with ≥2 members and **zero** AFDB hit (`is_novel_family`). Then explicitly fold one rep per family with ESMFold and re-confirm novelty with real coordinates (B1a).
- **Tools:** FoldSeek (commit `8dc75c74…`) for clustering; **ESMFold = facebook/esmfold_v1** (transformers 4.40.2, torch 2.2.2, python 3.10, env `esmfold`).
- **Commands:**

```
$FS cluster merged clu tmp --threads 32 --split-memory-limit 400G \
    --single-step-clustering --cluster-mode 2 -s 4          #
    ↪  62b_tier1_foldcluster_fast.sh
# ESMFold (single-seq, no MSA):
↪  EsmForProteinFolding.from_pretrained("facebook/esmfold_v1")
# reconfirm: $FS easy-search <pdb> <PDB/AFDB> --alignment-type 1 ... (confirmed-novel if
↪  no TM>=0.5 hit)
```

- **Outputs / counts:**
  - tier1 `headline.txt`: **NOVEL-FOLD FAMILIES (≥2, 0 AFDB) = 68,808** (`awk '$8==1' family_summary.tsv | wc -l` = 68,808); 169,651 proteins in them. tier2 adds 156,743.
  - ESMFold reps extracted: 68,808 (`extract_47783326.out`). **Structures actually built = 68,527** (281 failed to fold; `wc -l nf_queryDB.index` = 68,527), stored `06_esmfold_novelfold/pdb.tar.zst`.
- **Caveats:** ⚠ **68,808 = strict novel-fold (tier1) families**; tier1+tier2 = 225,551 (≠ 334,810, which is the unrelated sequence count). The explicit ESMFold *folding* job

for the 68,808 reps appears to have run off-repo (likely ORIX) and was imported as `pdb.tar.zst` — NOT FOUND as an in-repo array script.

3.13 Part A — A1 atmospheric "life on air" ⚠ (DIAMOND, not HMMER)

- **Purpose:** confirm EQ hydB/coxL hits are bona-fide high-affinity atmospheric enzymes (Group 1h/1l [NiFe]-hydrogenase; form-I CO dehydrogenase) — defeat the "group 1h is a wastebasket" objection.
- **Tools: ⚠ DIAMOND 2.0.15 best-hit vs reference DBs — NOT HMMER/HMM profiles.** HydDB for hydB; 3 RefSeq form-I refs + the `AYRGAGR` active-site motif for coxL. Trees: MMseqs2 18.8cc5c (derep) → **MAFFT v7.526 –auto → FastTree 2.2.0** (⚠ not IQ-TREE).
- **Env:** mixed hardcoded binary paths (`activate eggnog` for diamond; mafft/FastTree from the `busco` env).
- **Commands** (`82a_atmo_refs.sh`, `82c_atmo_classify.sh`):

```
git clone --depth 1 https://github.com/GreeningLab/HydDB ; diamond makedb --in
↪  HydDB_all_hydrogenases.faa --db hyddb
diamond blastp -q hydB.faa -d hyddb --ultra-sensitive --max-target-seqs 1 -o
↪  hydB_diamond.tsv   # class = 3rd pipe-field
diamond blastp -q coxL.faa -d coxL_formI_refs --ultra-sensitive --max-target-seqs 1 -o
↪  coxL_diamond.tsv
# coxL form I if (pident>55 & qcov>=90) OR motif AYRGAGR ; trees: mafft --auto | FastTree
```

- **Outputs / counts** (`$REPO/results/atmospheric/`):
  - hydB HydDB: **1l 5,292 prot / 2,609 MAG**; **1h 986 / 893** → 3,386 MAGs (22%) high-affinity.
  - coxL: **form_I 5,645 / 3,828** (25%); ambiguous 8,003 (NOT counted as CO oxidisers).
  - **Guild core (both H_2 + CO) = 1,204 MAGs** (`wc -l guild_core_mags.txt`).
- **Caveats: ⚠ Tree-label quirk** — `82c` concatenates the coxL form-I refs to **both** `hydB.nwk` and `coxL.nwk`, so `hydB.nwk` contains 3 CO-dehydrogenase refs and **no HydDB group refs** (HydDB groups live in `hydB_class.tsv`, not the tree). Fix before using `hydB.nwk` as a publication figure. **Honest negatives:** trace-gas enzymes are the *known* machinery (novelty = prevalence); the dark proteome does **not** supply trace-gas enzymes (corrected operon figure 177/11,923 = 1.5%); Group 1l absent from all 1,221 culture genomes.

3.14 Part A — A2 UV/desiccation repair

- **Purpose:** census NER/BER/NHEJ/photolyase/SOS/etc. across 15,469 MAGs; test light-defence vs sun exposure.

- **Tool:** awk/grep over existing eggNOG `.emapper.annotations` (KEGG_ko col 12) — a **28-KO panel**. No new search.
- **Command** (`$WB/scripts/ibex/124_uv_survival_census.sh`): per-MAG KO match against the inline panel.
- **Outputs / counts** (`$REPO/results/uv_survival/group_prevalence.tsv`):
  - BER **99.5%** (15,398); NER **99.4%** (15,382); SOS 94.2%; oxidative 91.9%; **NHEJ 78.4%** (12,122); photoprotective pigment 58.5%; MMR 48.8%; photoreactivation 17.5%.
  - Compartment: photolyase **Surface 22.4% / Deep 13.1%** (Surface>Deep MWU p=1.1e-14); NHEJ runs opposite (S 73.0 < D 81.1).
- **Caveats:** KO-only — splB, 6-4 photolyase, MAA/scytonemin not cleanly KO-resolvable (HMM follow-up flagged, not done).

3.15 Part A — A3 osmolytes (dark-family vignette)

- **Purpose:** profile dark families decoding to **osmolyte phosphatases** (membership, dN/dS, taxon, compartment TPM, plant host, salinity partial-correlation). A *vignette* within the dark-matter ecology follow-up, **not** a catalogue-wide trehalose/ectoine census.
- **Tool:** local python (`$REPO/code/phase_e_partial_osmolyte.py`; numpy/scipy). EC set = trehalose-6-P phosphatase (3.1.3.12), glucosylglycerol-P phosphatase (3.1.3.69), mannosyl-3-P-glycerate phosphatase (3.1.3.70).
- **Outputs:** `$REPO/results/darkmatter_ecology/indicators/partial_*.tsv`, `OSMOLYTE.md`.
- **Catalogue-wide osmolyte markers** (from A-general census, §3.18): otsA 12,914 prot/10,489 MAG, otsB 11,436/9,938, ectC 2,899/2,625, ectD 3,246/2,759.
- **Caveats:** ⚠ local non-Slurm script; env not declared. It reads GTDB from `$REPO/workbench/results/soil_gtdbtk/` — a **different** (broken-symlink) path than A1's `02_taxonomy/gtdbtk_out/`. ectoine modelled in the BGC census (A4), not as a dark family here.

3.16 Part A — A4 BGCs + rhodopsins + (alt-nitrogenase NULL)

- **Tool: antiSMASH 8.0.4** (IBEX module; `--minimal --genefinding-tool prodigal`). Rhodopsin/nitrogen = awk over eggNOG.
- **Command** (`$WB/scripts/86_antismash_array.sh`, `--array=1-774%60`, BATCH 20):

```
antismash --minimal --genefinding-tool prodigal --cpus $N --output-dir $ODIR
↪  --output-basename $mag $fna
```

  Aggregated by `aggregate_mobilome.py` (parses `*.region*.gbk`, joins GTDB + sample metadata).
- **Outputs / counts** (`$REPO/results/bgc_*`):
  - bgc_per_mag = **15,226** MAGs with ≥1 region (243 produced no output).

  - Top products (BGCs/MAGs): terpene 19,320/9,984; NRPS 18,078/3,914; T3PKS 7,983/6,878; RiPP-like 7,750/4,900; NI-siderophore 4,038/2,868; **ectoine 2,923/2,626**.
  - Rhodopsins: 398 hits / **341 MAGs** (`06_metabolism/rhodopsin/`; not yet copied to `$REPO`).
- **Caveats:** ⚠ **`--minimal` was deliberate** (no MIBiG/ClusterBlast matching) — product *types* reliable, but **no novelty/known-cluster comparison** (the A4-novelty step is an open task). ⚠ **Alt-nitrogenase NULL:** `07_nitrogen/` is **empty**; no vnf/anf script; vnfD (K22896) = **0 hits**; Mo-nitrogenase near-absent (nifH 25/20). The "V-nitrogenase × XRF vanadium" idea is unsupported.

3.17 Part A — A5 resistome / biosafety

- **AMRFinderPlus 4.2.7** (DB **2026-05-15.1**; ⚠ unpinned in `90_setup_mobilome_envs.sh` — `amrfinder -u`):

```
amrfinder -p $faa --plus --name $mag --threads $N -o $OUT     # 94_amrfinder_array.sh
```

  - β-lactam **4,873 hits / 3,816 MAGs = 24.67%** (`results/amr_class_summary.tsv`); aminoglycoside 2,002 MAGs; phenicol 1,108; rifamycin 1,086; macrolide 705; glycopeptide 622.
  - STRESS/metal: arsenic **2,423 MAGs**, mercury **1,077**. As-load vs salinity proxy Spearman **r=+0.260, p=3.1e-5, n=1,059** (⚠ XRF panel has no As/Hg → geochemical-stress link, not cognate).
  - Job: AMRFinder array `amrfind_47543961_*` (⚠ inferred from log names, not README-confirmed).
- **VFDB** (mmseqs2 `easy-search` id≥0.4 c0.7 e1e-10, module mmseqs2/14.7; query = 5,438 dRep-rep proteomes):
  - raw `vfdb_hits.tsv` = **1,267,855 lines / 346 MiB**; MAGs with ≥1 hit collapse 5,435 (id≥40) → 202 (id≥90); exotoxin proteins 6,260 → 16 (id≥90). "Virulence" is dual-use; collapses with stringency.
- **Pathogen census** (local python `pathogen_census.py` over GTDB summaries; no Slurm):
  - **7/15,469 named pathogen species = 0.05%** (E. coli 5, A. baumannii 2); genus-level human/animal 126 MAGs (0.8%) but overwhelmingly environmental relatives ⇒ **clean, low-anthropogenic biome** (contamination control passes).

3.18 Part A — A-general marker census (eggNOG KO → GTDB phylum)

- **Tool:** awk over eggNOG (`82_metabolism_marker_census.sh`); by-phylum join in `aggregate_mobilome.py` (reads `02_taxonomy/gtdbtk_out/`).
- **Outputs / counts** (`marker_summary.tsv`): coxL(K03520) 13,675 prot/5,801 MAG; hydB(K06281) 6,602/3,665; rbcL 3,892/3,229; nifH 25/20; otsA 12,914/10,489; otsB 11,436/9,938; ectC 2,899/2,625; recA 12,839/11,302. Per-phylum tables

metabolism_{coxL,hydB,rbcL,nifH}_by_phylum.tsv.

- **Caveats:** raw KO census **cannot** resolve hydrogenase subgroup — that is what A1 (DIAMOND/HydDB) does. Use A1's form-I coxL (5,645), not the raw KO 13,675, for "CO oxidisers".

3.19 Part B — B2 selection (dN/dS) ⚠ (Biopython NG86, not codeml)

- **Purpose:** are dark families under purifying selection (dN/dS≪1 ⇒ real genes)?
- **Tool:** ⚠ **Biopython `cal_dn_ds(method="NG86")` pairwise screen** (MAFFT codon alignment). **Not codeml, not HyPhy** (HyPhy is B4 only). Env `eco_probes`; mafft from `busco` env.
- **Command** (`code/ibex/94_darkfam_dnds.sh`, `--array=0-N%30`, 200 families/task): per-pair `cal_dn_ds`; filters ≥30 codons, 0<dS≤2 (saturation), ≤80 pairs/family, ≥3-member families.
- **Outputs / counts** (`…/selection/dnds.tsv`, repo copy `results/darkmatter_ecology/dnds.tsv`):
  - DARK median **0.391** (n=22,767, frac<1 = 0.993) vs CONTROL **0.286** (n=21,579); MWU p≈0.
  - Endemic (≤3 sites) **0.332** vs EQ-wide (≥30) **0.420**, **MWU p=5.3e-69**.
- **Caveats:** NG86 pairwise is a **screen, not a codon-model test**. Structured-vs-cosmopolitan difference (0.389 vs 0.392) is tiny though significant — the "adaptive" signal is weak.

3.20 Part B — B3 ecology (CoverM + bespoke PERMANOVA) ⚠ (not vegan)

- **Purpose:** is the dark proteome environmentally structured + endemic, not random?
- **Tools:** abundance = **CoverM `contig -p minimap2-sr`** (551 samples). Stats = ⚠ **bespoke numpy/scipy reimplementation `code/eco_stats.py` (McArdle-Anderson PERMANOVA, Borcard varpart, Mantel) — NOT vegan / NOT scikit-bio.** Hellinger + Bray-Curtis. Presence = covered_fraction ≥ 0.5.
- **Commands:** CoverM `code/ibex/91_darkfam_coverm_array.sh` (`--array=0-550%30`); stats `code/phase_b_ecology.py`, endemism `code/phase_d_endemism.py`.
- **Outputs / counts:**
  - Abundance matrix `…/darkfam_abundance.tsv` (92 MB, 23,382 fam × 551). Working universe = **23,382 multi-member (≥3) dark families** (`CATALOG_STATS.txt`).
  - PERMANOVA (all p=0.001 except V): compartment R^2=0.020; space R^2=0.038; climate R^2=0.047; geochem R^2=0.066; ⚠ **vanadium R^2=0.004 p=0.325 (n.s.)** — the XRF-V signal does NOT survive.
  - Endemism (`ENDEMISM.md`): single-site **543 (2.3%)**; ≥30-site EQ-wide **10,669 (45.6%)**; median 26 sites.
  - Geochem partial correlations (control lat+lon, q<0.05): S 79% retained, Cl 84%, Ca 72%, salinity 57% — environmental, not geographic.

- **Caveats:** R^2 are small (≤0.066) but significant. The bespoke stats should be validated vs real vegan if a reviewer asks. "Plant-host specificity" = the `compartment` factor.

3.21 Part B — B3d abundance (not a contamination tail)

- **Tool:** local python `code/darkmatter_abundance.py` over the CoverM TPM matrix.
- **Outputs / counts** (`results/darkmatter_ecology/darkfam_abundance_stats.tsv`, 23,382 rows): median peak **584.75 (~585)**; median prevalence **52 samples**; single-sample (≤1) only **1.3%**; **peak ≥10 = 99.5%**; 89.4% peak ≥100.
- **Caveats:** abundant where present, broadly detected — but only 7.9% are *both* prevalent (≥50% of samples) *and* non-trace; patchy (consistent with endemism), emphatically not a rare/dead-DNA tail.

3.22 Part B — B4 diversifying selection + Boltz-2

- **Tools:** NG86 screen → the dN/dS>1 tail; per-site **HyPhy FUBAR** (MAFFT→FastTree→`hyphy fubar`); structure ESMFold→FoldSeek→p2rank pockets; ligand **Boltz-2** (single-seq, ORIX H200). Versions ⚠ NOT FOUND.
- **Outputs / counts** (`results/darkmatter_ecology/positive_selection/`):
  - **148 families** dN/dS>1 (0.65% of 22,767); 123 robust; median calibrated P 0.22.
  - Boltz-2: dark **lectin 58PRr3** best binds **GlcNAc/NAG** (ligand_iptm 0.55, binding_prob 0.48).
- **Caveats:** ⚠ honest negative — of 34 nominal positive sites in the lectin, **0** fall in the predicted NAG-contact site (≈chance); interface-localised selection unconfirmed. Decode-able families are shallow (3–4 members) = exactly where per-site selection has no power.

3.23 Part B — B5 decode (C·V·T) ⚠ (dossier = 73, not 10)

- **Purpose:** assign candidate EC/reaction to dark families via the in-house **C·V·T (context·verify·tier)** engine; calibrate P(correct) against a held-out (LRO) set.
- **Tools:** in-house C·V·T scoring + `gapsmith` gap engine. Decode-join = awk (`code/ibex/97_darkfam_decode_join.sh`).
- **Inputs:** C·V·T score table `06_darkmatter/scores_v3/candidate_scores_v3.tsv` (**11.4 GB**); LRO calibration `calibration_{pilot,sub}/reliability.tsv`.
- **Outputs / counts:**
  - `family_ec.tsv` (repo copy in `results/darkmatter_ecology/`): 23,382 families, **23,083** with a specific EC.
  - Dossier `06_darkmatter/dossier/index.tsv` = **73 candidates**; the **"10 dark enzymes" = the top-10 marquee rows**, calibrated P (`calP`) ≈ **0.24–0.45** (e.g. 2.3.1.8 phosphate acetyltransferase 0.445; 2.6.1.1 aspartate aminotransferase 0.380).

- **Caveats:** ⚠ `results/darkmatter_function_assignment/` **does NOT exist** — decode artefacts live under `06_darkmatter/{scores_v3,dossier,calibration_*}`. ⚠ `tools/gapsmith_src` referenced but **not located** on disk. All decode calls are explicitly **low-confidence hypotheses** (calibrated P ≈ 0.22–0.48), not annotations.

3.24 Part C — C1 PICRUSt2 reference bias

- **Purpose:** is gene content predictable from 16S? PICRUSt2 KO (amplicon companion) vs MAG-encoded KO, same samples.
- **Tools:** PICRUSt2 **v2.4.1** (from the amplicon repo, not re-run). MAG side: eggNOG KO + CoverM relab; build (`scripts/130_mag_ko.sbatch` → `130_mag_ko_metagenome.py`). ⚠ path is `scripts/`, not `scripts/ibex/`.
- **Outputs / counts** (`results/picrust2_bias/SUMMARY.md`): **436** matched samples; weighted NSTI median **0.159** (placement fine); presence-recall **93.3%**; predicted **10,543** vs encoded **4,482** KO/sample (**2.4×**); **56.7%** of MAG proteins have **no KO** (19.1% no eggNOG ortholog). Marquee FPs: hoxH 99.1% predicted / 0.2% encoded; anfG 99.1% / 0.0%.
- **Caveats:** placement is NOT the failure (NSTI fine); the failure is over-prediction + decoupled abundance + the KO-space ceiling. Reinforces the dark-matter pillar.

3.25 Part C — C1a pangenome openness ⚠ (ran 2026-06-28; scipy missing)

- **Purpose:** *why* 16S→function fails — are EQ species' pangenomes open (accessory unpredictable)?
- **Tool:** mmseqs2 (`eggnog` env), Heaps'-law in `scripts/ibex/140_pangenome_openness.py` (`--min-seq-id 0.5 -c 0.8`).
- **Outputs / counts** (`06_pangenome/out/enrichment_summary.json`): **345 species, 8,635 genomes; 337 open / 8 closed**; median Heaps α 0.555; **dark OR accessory-vs-core = 1.786** (accessory dark-frac 0.082 > core 0.048).
- **Caveats:** ⚠ status was "PLANNED" in the outline but **it has now run** (outputs dated 2026-06-28). ⚠ `"scipy": "unavailable"` ⇒ the Fisher p on the dark core-vs-accessory 2×2 was **not computed** — a 1-line gap to close.

3.26 Part C — C2 SMETANA community (16-community pilot) ⚠ (detailed deferred)

- **Tools:** gapseq **v1.4** GEMs (env `gapseq-env`); **SMETANA** global MRO + FBA cross-feeding (cobrapy + GLPK, env `metabolism-env`).
- **Commands:** `gapseq doall <MAG>.fna` (`125_…array.sh`, 316 GEMs); `smetana --flavor fbc2 --global/--detailed` (`126_…run.sh`, 16 communities); FBA `129_fba_crossfeed.py`.
- **Outputs / counts:**

  - SMETANA global: 16 communities, **MRO median ≈ 0.69** (range 0.55–0.78); MIP = n/a everywhere.
  - FBA cross-feed: **84,106** organic donor→receiver edges. Guild test (316 MAGs, 27 guild): donor potential guild 232 vs non-guild 243, MWU **p=0.694** ⇒ **guild-as-feeder REFUTED**; completeness control |ρ|<0.07.
- **Caveats:** ⚠ **No CPLEX/Gurobi on the cluster** (access-restricted) → SMETANA **detailed** (MILP) intractable with GLPK → **deferred** (probe scripts `cplex_*.sbatch` present). Complete-medium FBA = potential *upper bound*. Pilot = 16/547 samples.

3.27 Part C — C3 global / desert specificity ⚠ (SPIRE not landed)

- **Tool:** DIAMOND blastp `--sensitive -e 1e-3 -k 1` of the 23,382 dark-family reps vs each catalogue (env `eggnog`).
- **Commands:** `131_global_specificity.sh` (GEM/SPIRE); `131b_desert_specificity.sh` (5 deserts).
- **DB paths:** `databases/{gem,spire,deserts}/`; GEM = 52,515 MAGs (`gem_all.faa`, 26.7 GB); deserts = SPIRE per-study genecalls (Atacama, Mojave, arid, trace-gas-autotrophy, Antarctic).
- **Outputs / counts:**
  - **GEM 47.4% absent** (11,075/23,382); **MGnify90 36.6%** (prior run); deserts: Atacama 89.1%, autotroph 62.8%, Antarctic 55.9%, Mojave 51.9%, arid 41.7%; **desert-union 31.9%**.
  - **Intersection (MGnify ∪ GEM ∪ all-deserts): 5,011/23,382 = 21.4%** with no homolog anywhere.
  - Endemism **plateaus at ~32–47%** at the deepest references (not →0) = genuine endemism, not undersampling.
- **Caveats:** ⚠ **SPIRE search not landed** — no SPIRE row in the summary TSV; the **21.4% is provisional (≤21.4%, tightens when SPIRE completes)**. Endemic families are under stronger purifying selection than shared ones.

---

4. Three pillars → results map

Locations are under `results/` (paper repo) unless marked **WB** (workbench); "fig" = the matching `fig_*` figure. The `verify.py` claim ID for each beat is in `reproducibility/REPORT.md`.

Beat	Analysis	Key result	Location
A0	GTDB-Tk	44 phyla, 0 novel; Actino 6,788	WB gtdbtk_out
A1	HydDB + trees	guild core 1,204	atmospheric; fig lifeonair

Beat	Analysis	Key result	Location
A2	eggNOG 28-KO	NER 99.4 / NHEJ 78.4	uv_survival
A3	osmolyte vignette	osmolyte phosphatase families	darkmatter_ecology / indicators; OSMOLYTE.md
A4	antiSMASH	terpene 64.6%, ectoine 17%	bgc; fig bgc
A5	resistome census	β-lactam 24.7%, pathogens 0.05%	amr, vfdb, pathogens; fig amr
B1	dark → fold ladder	2.53M struct-novel; 68,808 folds	WB 04_structure
B1a	ESMFold	68,527 structures	WB 06_esm-fold_novelfold
B2	dN/dS (NG86)	dark 0.39 ≪ 1	darkmatter_ecology / dnds.tsv; fig dnds
B3	ecology	endemism 45.6 / 2.3	darkmatter_ecology; fig pcoa / permanova
B3d	abundance	median peak 585	darkfam_abundance.tsv (large)
B4	FUBAR + Boltz-2	dark lectin → GlcNAc	positive_selection
B5	C·V·T decode	top-10 of 73-cand dossier	family_ec.tsv
C1	PICRUSt2 vs MAG	2.4×; 56.7% no-KO	picrust2_bias; fig picrust2
C1a	pangenome	337 / 345 open	WB 06_pangenome
C2	gapseq + SMETANA	MRO 0.69; guild refuted	smetana
C3	vs catalogues	21.4% absent everywhere	global_specificity; fig global

5. Reproducibility & setup gaps

`bash reproducibility/run_all.sh` regenerates 11 figures, recomputes the tracked claims via `reproducibility/verify.py`, and writes `REPORT.md` (`REPRODUCIBILITY: OK` on success). On a **fresh checkout** two documented setup steps must run first, or 7 COMPUTE claims fail with "No such file or directory":

1. **Large files** — `reproducibility/fetch_large_files.sh` rsyncs 7 large inputs (88 MB abun-

dance table, etc.) from the IBEX staging dir …/EmptyQuarter_Data/processed/mags-paper-large-files/. ⚠ The published script uses **SSH** (off-cluster); **on IBEX the staging dir is a local path** — copy it directly instead.

2. **The `workbench` symlink** — currently dangling (→ `/home/leechuck/…`); repoint to `rob/metagenomics`.
3. **Taxonomy path** — `verify.py` expects `workbench/results/soil_gtdbtk/gtdbtk.{bac120,ar53}.summary.ts` but the canonical file is `workbench/02_taxonomy/gtdbtk_out/…` (matches the paper exactly). Reconcile with a symlink or a 1-line path edit.

After those, expect `COMPUTE: 37 pass, 0 fail.` `verify.py` tiers: COMPUTE (recomputed), TABLE (recorded statistic), PROVENANCE (heavy-pipeline output, documented).

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6. Parked / not-adopted tracks (so nothing is mistaken for the main path)

- **DAS_Tool refinement** — DAS_Tool builds a **consensus of the two SemiBin2 binnings** (`-i single.contigs2bin.tsv,self.contigs2bin.tsv -l single,self`), piloted on candidate-novel PlantRoot samples then scaled to 542 (527 produced bins; 4 provenance-failed, backfilled 2026-06-07). Result: 14,426 refined bins → dRep → **5,438 reps**. **Not adopted:** the diagnostic (`diag_dastool_f47.sh`) showed consensus bins mostly score ~0 on DAS_Tool's SCG score, and the catalogue (15,469, `__SemiBin_` IDs) is single_sample SemiBin2 — **DAS_Tool bins were never the annotation substrate.** Likely low value here because both inputs are the *same binner* (DAS_Tool gains most from diverse binners).
- **self_supervised binning** (21,616 HQ+MQ) — an enrichment pool feeding the single-vs-self dRep comparison (9,211 reps), not the headline census.
- **delta protein catalogue** (pyrodigal, 414M reps) — a larger assembly-wide catalogue parallel to cluster90; feeds tier1/tier2 fold clustering.
- **SPIRE** in C3 — search not completed; the 21.4% homolog-free figure is provisional (≤21.4%).
- **07_nitrogen / vnf-anf phylogeny** — never populated (vnfD = 0 hits); the vanadium-nitrogenase angle is a clean null.
- **A4 BGC novelty** (BiG-SCAPE/MIBiG) — `--minimal` antiSMASH gives product types only; the novelty comparison is an open task.

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7. The catalogue/binning issue & remediation plan ⚠⚠ (read this)

This is the single most consequential finding in the pipeline. It is summarised in §3.3 (binning mode) and §3.6 (dRep lineage); this section consolidates **what exists, what is missing, and how to fix it** so the decision can be made without re-deriving it.

7.1 The issue in one paragraph

The project's **authoritative binning is self-supervised** — the Snakemake at `alelopezv/EQ_WGS` runs `SemiBin2 single_easy_bin --self-supervised -m 1000` (BWA-MEM coverage) → `wgs_assemblies/binning/<sample>/semibin/output_bins/`. But the **executed catalogue used a separate PRETRAINED re-binning** Robert ran (`01_semibin2_single.sh`, `--environment global`, minimap2, no `model.pt`). The 15,469-MAG catalogue (`03_annotation/mag_list.txt`) is entirely that pretrained `single_sample` track, and the whole downstream pipeline (proteins → dark matter → GTDB headline → Part A/B/C) was wired to it. The self-supervised bins were fully QC'd and annotated but only fed the unadopted `04_drep_self_vs_single` comparison.

7.2 What you actually have: a best-of-both MERGE

`04_drep_self_vs_single/out/dereplicated_genomes/` = **9,211 species representatives**, dereplicated at 95% ANI from the pooled **37,085** bins (15,469 pretrained-single + 21,616 self-sup). dRep kept the single best representative per species cluster regardless of track:

```
6,308 reps = self-sup   |   2,903 reps = single(pretrained)      (Wdb.csv)
```

So **self-sup supplies 68% of the merged catalogue** — it is the scientifically strongest genome set on disk. (Per sample the self-sup track also recovers more good bins, e.g. 59PRr1: 81 self-sup HQ+MQ vs 66 single_sample.)

How much do the two tracks overlap? (cluster-level analysis of `Cdb.csv`, 9,211 species):

```
BOTH tracks recovered it (agree):   5,560   (60%)
self-sup ONLY (pretrained missed):  3,043   (33%)
pretrained ONLY (self-sup missed):    608   ( 7%)
```

⇒ **self-sup nearly subsumes pretrained**: self-sup covers 8,603 species (93%), pretrained 6,168 (67%); pretrained adds only **608 unique species (7%)**. The driver of the track difference is, in order:

1. ⚠ **min-len — the dominant driver (verified).** Self-sup forced `-m 1000`; pretrained left `--min-len` unset → SemiBin2's adaptive `ratio=0.05` rule (`utils.py::load_fasta`: `min_len = 1000 if bp(1000–2500)/total_bp < 0.05 else 2500`). Because the assemblies were built at `--min-contig-len 500` (fragmented soil), the 1000–2500 bp fraction is **25–31%** of bp — **5–6× the 0.05 cutoff** — so the rule resolves to **2500 bp for every assembly** (verified **20/20** evenly-sampled; the margin guarantees all 572). **The pretrained track therefore discarded ~27% of each assembly's bp (all 1000–2500 bp**

contigs) before binning; the self-sup track kept it. This alone plausibly explains the 3,043 self-sup-only species.

2. The **model** (self-supervised trained per-sample vs pretrained `global`).
3. Minor — self-sup's BAM included SE reads. The **coverage mapper (BWA-MEM vs minimap2-sr) is second-order** for binning and not a meaningful driver. Note: the project's *own* Snakemake dRep (`wgs_assemblies/binning/dereplication/`, config comp75/con25/Sani0.9) **never ran** — the only dedup combining both tracks is Robert's `04_drep_self_vs_single` (`-pa 0.9 -sa 0.95 -comp 50 -con 10`).

Housekeeping (disk): the self-supervised bins are **stored twice** — uncompressed source `wgs_assemblies/binning/<sample>/semibin/output_bins/*.fa` (~404 M/sample for 11Dr3) **and** a gzipped hard-copy `…/processed/…/<sample>/wgs/binning/self_supervised/output_bins/*.fa.gz` (~115 M/sample; verified a real gzip copy, not a symlink). Across ~570 samples this is on the order of **hundreds of GB** duplicated. Recommended cleanup: pick one canonical location (the Snakemake `wgs_assemblies/…/semibin/`), gzip it, and replace the processed copies with **symlinks** — but Robert's `mag_list_self_supervised.txt` + self_supervised annotation/GTDB inputs reference the processed `.fa.gz` paths, so symlink (don't delete) and confirm the canonical location with Robert first.

7.3 What the merge HAS vs what is MISSING

Layer	Status	Note
Bins (best-of-both)	**HAVE**	9,211 reps, `04_drep_self_vs_single/out/dereplicated_ge`
Per-bin CheckM2 QC	**HAVE**	both tracks
Per-bin GTDB-Tk	**HAVE** (split)	single → `02_taxonomy/gtdbtk_out/` (15,469); self-sup → `02_taxonomy_self_supervised/gtdbtk_out_chunks/` (21,616); 15,469+21,616 = 37,085 = dRep input, so **every rep is classified**
Per-bin Bakta + eggNOG	**HAVE**	both tracks fully annotated (Bakta `.faa` + eggNOG `.emapper.annotations`)
Consolidated GTDB summary (9,211)	**MISSING**	trivial join/subset of the two existing summaries by rep ID

Layer	Status	Note
Pooled protein catalogue (9,211)	**MISSING**	`mmseqs linclust` of the existing per-bin `.faa`
UniRef50 → ProstT5/ESMFold/FoldSeek dark-matter ladder	**MISSING**	was run on `single_sample` only — the heavy GPU re-run
CoverM abundance + Part A/B/C analyses on the 9,211	**MISSING**	scripts keyed to `single_sample` MAG IDs

One line: all expensive *per-bin* work (binning, QC, GTDB, Bakta, eggNOG) is **already done for both tracks**; only the **consolidated/downstream** layers keyed to the 9,211 are missing.

7.4 How to fix it — staged plan (do the cheap, decision-informing steps first)

Phase 1 — consolidate (cheap, hours, low-risk, only produces new derived tables):

1. Freeze the 9,211 rep list + per-rep track provenance from `Wdb.csv`.
2. Subset/join the two GTDB summaries → one 9,211-rep taxonomy table → **recount phyla**. (The headline 44 / 6,788 / 3,740 *will* change — this immediately shows how much the genome-level story moves.)
3. Pool the 9,211 reps' existing Bakta `.faa` → `mmseqs linclust` → new protein-rep set + count.

Phase 2 — assess before committing (cheap, the decision gate):

4. UniRef50 search on the new reps (CPU array) → new "dark" count; compare to 2.84M. If the dark fraction/composition is materially different, the full re-run is justified; if near-identical, robustness can be argued.

Phase 3 — the expensive re-run (GPU, weeks — only if Phase 2 warrants it):

5. ProstT5 + ESMFold + FoldSeek structure ladder on the new dark reps → novel folds.
6. CoverM abundance + re-run the (parameterized) Part A/B/C scripts on the new catalogue.

Recommendation: this is a real scientific + compute decision → **Robert's call.** The defensible path is **Phase 1→2 now** (cheap; quantifies the impact), then decide Phase 3 with data in hand. Keeping the pretrained `single_sample` catalogue and merely justifying it is hard to defend given (a) the intent was self-sup and (b) self-sup demonstrably recovers more/better bins and won 68% of dereplication.

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8. Glossary & naming conventions

- **MAG ID** `<sample>__SemiBin_<n>` — `__SemiBin_` ⇒ a SemiBin2 bin.
- **Sample ID** `<site><compartment><trip>` — e.g. `11PRr3` = site 11, PlantRoot, trip r3. Compartments: `S/D` Surface/Deep, `PR` PlantRoot; trips `r1–r3` (+ `F` re-runs).
- **HQ / MQ / LQ** — CheckM2 quality: HQ ≥90% comp / ≤5% contam; MQ ≥50/≤10; LQ below.
- **dark (sequence)** — a cluster90 rep with no UniRef50 hit (e<1e-3). **dark (structural)** — a dark rep with no AFDB FoldSeek hit. **dark family** — an orthogroup of ≥3 sequence-dark members.
- **novel family (334,810)** — multi-member cluster at 30% sequence identity. **novel-fold family (68,808)** — fold cluster ≥2 members, 0 AFDB hit (`is_novel_family`). *Different analyses; never summed.*
- **guild core** — MAGs encoding both atmospheric H_2 (Group 1h/1l) and form-I CO oxidation (1,204).
- **C·V·T** — the in-house context·verify·tier dark-function decode engine.
- **ORIX** — external H200 GPU cluster used as free-cycle overflow for ProstT5/ESMFold/Boltz-2.

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*Generated 2026-06-28 by forensic reconstruction from the IBEX workbench (`rob/metagenomics`) and the paper repo. Every count was verified against on-disk artefacts; ⚠ marks a flagged discrepancy or unverified item. The single most important open issue is the binning/catalogue provenance — **see §7 for the full state (what exists, what is missing) and the staged remediation plan**; confirm with Robert.*